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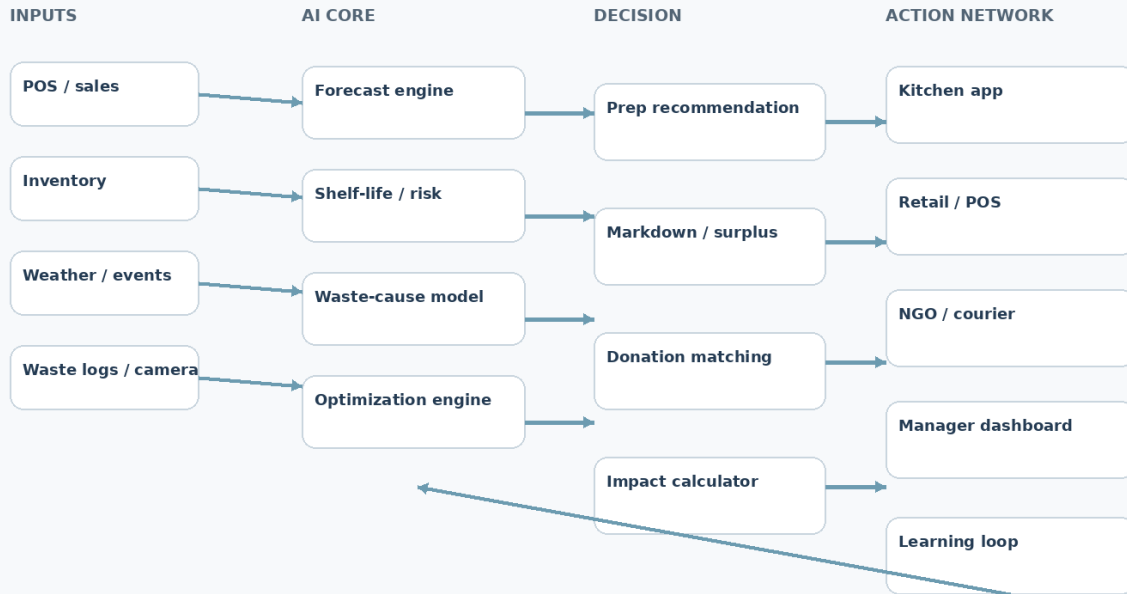
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SIH 2026 • PRE-BUILD RESEARCH DOSSIER

S11 — Closed-Loop Food Waste Intelligence Platform

From demand prediction → smarter preparation → waste intelligence
→ surplus recovery

Closed-Loop Food Intelligence Architecture



Every action becomes new training signal → forecast → better decision

How to use this report

Read Sections 4–8 before prototyping. They define the product wedge, MVP, architecture, data model, UI flow, and the exact demo story the judges should see. Sections 9–12 are for scalability, feasibility, competition strategy, and future expansion.

0. Statement verification note Back to TOC

Publicly indexed sources do not expose a reliable 2026 mapping for the shorthand “S11”. A third-party copy titled “SIH 2026 Problem Statements Overview” is indexed, but its visible text contains the older SIH 2025 identifiers (for example SIH25001–SIH25016). [0] Therefore, this report treats “S11” as the team-locked Food Waste Optimization problem from our working context, rather than claiming that the exact official 2026 wording has been independently verified on the SIH portal.

Action before final submission

Paste the official S11 title/background into this section when you have the SIH portal screenshot/PDF. The product architecture below remains valid for a food-waste optimization brief, but the final PPT must quote the official wording exactly.

1. Executive brief Back to TOC

The strongest version of this project is not “an app that tracks food waste.” It is a digital operating layer for food businesses that predicts what they are likely to need, recommends how much to prepare/order, detects why waste is happening, then routes unavoidable surplus to the highest-value destination—sell, donate, or recover—while feeding outcomes back into the next forecast.

Dimension	Recommended position
Problem	Food is wasted because demand, preparation, inventory, shelf life, and surplus decisions are disconnected.
Product	A closed-loop AI decision system for restaurants, hotels, cafeterias, supermarkets and institutional kitchens.
Core wedge	Prevention first: forecast demand + optimize prep. Rescue second: markdown/donate unavoidable surplus.
Differentiator	One learning loop across operational data, waste events and recovery outcomes—without forcing a new workflow.
MVP	CSV/POS ingestion, demand forecast, prep recommendation, waste logging, surplus routing, impact dashboard.
Scale	Multi-tenant SaaS + API-first integration layer + optional computer vision + local policy/configuration.

2. The problem in real numbers Back to TOC

Food waste is large enough to justify an infrastructure-style solution, not a one-off awareness campaign. UNEP estimates 1.05 billion tonnes of food were wasted in 2022 across households, food service and retail —about 19% of food available to consumers. Food service accounted for 290 million tonnes and retail 131 million tonnes. [1][2] FAO separately estimates 13.2% of food is lost after harvest and before retail, with another 19% wasted at retail, food service and household levels. Food loss and waste is associated with roughly 8–10% of global greenhouse-gas emissions. [3]

Why this matters for SIH		
The opportunity is not only environmental. Waste is also a business-systems problem: wasted food means wasted procurement spend, labor, refrigeration, transport, shelf space and disposal capacity. A system that prevents even a small percentage of avoidable waste can create measurable ROI for operators.		
Sector	2022 global waste (UNEP)	What this implies for the product
Households	631 Mt	Huge impact, but difficult as a first enterprise wedge.

Food service	290 Mt	Strong fit: recipes, prep volumes, service schedules and plate/prep waste create rich operational signals.
Retail	131 Mt	Strong fit: expiration, inventory, promotions, pricing and redistribution create optimization opportunities.

3. Working problem statement Back to TOC

Food businesses routinely make decisions with incomplete information: how many customers will arrive, which menu items will sell, how much to prep, what inventory is actually usable, what will expire first, whether a promotion will move surplus, and who can receive safe surplus before the disposal deadline. Existing systems often handle one layer well but leave the decision chain fragmented.

3.1 The root causes Back to TOC

- Demand uncertainty: weekdays, weather, holidays, events, promotions and local patterns can shift demand quickly.
- Operational friction: staff often have no time for detailed manual logging during peak service.
- Perishability: the same “1 unit” can have very different economic risk depending on shelf life and storage conditions.
- Disconnected tools: POS, inventory, procurement, kitchen production, waste logs, pricing and NGO coordination frequently sit in separate systems.
- Late intervention: many systems explain waste after it happens instead of changing the decision that caused it.
- No learning loop: an operator may record waste but not systematically feed that signal back into tomorrow’s prep plan.

4. The product idea — what we should actually build Back to TOC

Working concept: “NourishOS” (name can change). The system acts like an AI operations co-pilot for food businesses. It watches demand, inventory and waste signals; predicts risk; recommends the next best action; and measures the result.

4.1 The closed-loop cycle Back to TOC

1. Observe — ingest sales, inventory, menus, calendar, weather, promotions and waste events.
2. Predict — forecast demand by item/time window and estimate spoilage / surplus risk.
3. Decide — recommend production, procurement, stock rotation, markdown or redistribution.
4. Act — staff accept/override recommendations inside a simple workflow.
5. Recover — route safe surplus to customers, partners, food banks or other recovery channels.
6. Learn — compare forecast vs actual and recommendation vs outcome; update the next plan.

4.2 What makes it different Back to TOC

Layer	Typical market solution	Our integrated approach
Prediction	Forecasting/ordering software	Forecast + prep + waste-risk cost in the same decision.
Measurement	Waste tracker / smart bin / camera	Manual-first MVP; optional camera/scale later; every event becomes a learning signal.
Intervention	Static alerts or recommendations	Rank actions by financial + environmental impact and expected confidence.
Surplus	Marketplace or charity platform	Decision engine chooses sell vs donate vs recover based on time-to-expiry, demand and partner capacity.
Learning	Reports and dashboards	Closed loop: outcome of each action affects future recommendations.
Deployment	Enterprise systems	API-first, CSV-first, then connectors; designed for small business through multi-site enterprise.

5. Existing solutions — who is already doing what? Back to TOC

There is serious competition. That is good news: it proves the market need. The strategic mistake would be to claim that nobody is solving food waste. Instead, we should show that the market is fragmented across different moments of the waste lifecycle.

Player	Strongest capability	Current scope	Gap / opportunity for our product
Winnow	AI/touchless waste measurement in commercial kitchens	3,500+ sites across 94 countries; camera + scale + AI; insights on what/why is wasted. [4]	Excellent capture and kitchen analytics, but our differentiation can be the upstream planning + downstream redistribution loop.
Orbisk	AI vision + weighing + actionable kitchen recommendations	1,000 professional kitchens; automatic capture, ingredient-level insights and AI-powered actions. [5][6]	Strong prevention after measurement. Opportunity: combine with demand forecasting, retail/consumer surplus and cross-organization recovery.
Leanpath	Food-waste prevention and	AI tracking + prescriptive	Deep enterprise capability.

Player	Strongest capability	Current scope	Gap / opportunity for our product
	operational workflows	recommendations + alerts; used across restaurants, hospitals, universities and hotels. [7][8]	We should not copy it; instead differentiate on broader ecosystem orchestration and lower-friction deployment.
Too Good To Go	Surplus monetization and enterprise surplus management	Marketplace + platform for near-expiry inventory, AI markdowns and donation routing; 120M app users are cited on its platform page. [9]	Excellent last-mile demand. It begins when surplus already exists; our system should try to prevent that surplus before it is created.
FoodCloud	Charity/community redistribution	Connects food businesses with local charities/community groups; reports 450M meals redistributed and 3 continents reached. [10]	Strong social recovery layer. Our role can be upstream decisioning plus a neutral routing layer.
Wasteless	AI dynamic pricing / markdowns	Uses real-time inventory, freshness, price sensitivity, waste events and sell-out dynamics. [11]	Strong retail optimization. Opportunity: add production planning and foodservice workflows, and connect pricing decisions with donation/circular pathways.
Freshflow / Afresh	Demand forecasting + replenishment	Freshflow targets fresh produce ordering; Afresh covers demand forecasting, inventory and replenishment across grocery. [12][13]	Strong upstream optimization. Opportunity: connect forecast decisions to actual waste causes and recovery outcomes.

6. The strategic gap: what is still missing? Back to TOC

Based on the documented capabilities of these products, the clearest opportunity is not another isolated feature. It is an interoperable “food decision loop” that spans multiple stages. This is an inference from the current product landscape, not a claim that competitors could never offer such functionality. [4]–[13]

Fragment	What happens today	Our opportunity
Before production	Forecasting/order recommendations	Convert demand forecast into exact prep and procurement targets with uncertainty bands.
During operations	Waste is measured or manually	Capture the reason, stage and item

Fragment	What happens today	Our opportunity
	logged	with near-zero friction.
Near expiry	Markdown/marketplace tools activate	Choose the next best action using time, margin, safety rules and local demand.
Donation	Separate charity network	Match surplus to verified recipients by distance, capacity and pickup window.
After action	Report generated	Measure money saved + kg avoided + meals recovered + emissions avoided and learn from the outcome.

Our “moat” should be the graph, not a single model

The defensible asset is the connected operational dataset: item → forecast → prep → actual demand → waste event → reason → intervention → outcome. Over time, this lets the system learn which interventions work for which location, menu, season and customer pattern.

7. Real-world scenario — the story judges can instantly understand Back to TOC

Scenario: a hotel breakfast buffet in Jaipur Back to TOC

07:00 — The hotel historically serves 480 guests on Saturday, but demand is sensitive to local events and occupancy. The model sees 82% occupancy, a school-holiday weekend and a nearby convention ending late Friday.

07:10 — Instead of blindly preparing 550 portions, the system recommends: 510 baseline portions, 40 portions staged as a second-batch trigger, and smaller first-batch quantities for pancakes and cut fruit.

09:30 — Sales/consumption signals show pancakes are moving 24% slower than expected while fruit is moving faster. The system recommends stopping the next pancake batch and increasing fruit replenishment.

11:00 — The breakfast window ends with 42 safe portions of selected items. The system checks local recovery partners. One verified NGO can receive 30 portions with a pickup within the safe window; another 12 portions are offered through a local surplus marketplace. Remaining unavoidable scraps are classified for compost/organic recovery.

12:00 — The manager sees the result: 38% less avoidable prep waste than the same Saturday baseline, ₹4,800 estimated cost avoided, 42 portions recovered, and a new learning signal for next Saturday.

Why this scenario is strong

It demonstrates prevention, adaptation, recovery and measurable impact in one continuous story. It also avoids relying on a futuristic robot: the MVP can work with normal hotel/restaurant data plus a few simple human inputs.

8. MVP scope — what we should build first Back to TOC

Module	MVP feature	Demo priority
1. Organization	Business type, location, timezone, operating hours, units/currency	Must have
2. Data import	CSV upload for sales + inventory + menu; sample dataset for demo	Must have
3. Demand AI	7-day/next-service forecast, confidence band, explanation	Must have
4. Prep optimizer	Recommended quantity + “why” + manual override	Must have
5. Waste capture	1-tap/3-tap waste logging with item, quantity, reason, stage	Must have
6. Waste intelligence	Top drivers, cost, trend, root-cause clustering	Must have
7. Surplus router	Sell / donate / recover recommendation with deadline	Must have
8. Impact dashboard	kg, meals, cost, CO2e proxy, recovery rate	Must have
9. CV waste capture	Camera recognition + weight integration	Phase 2
10. External integrations	POS/ERP/marketplaces/NGOs	Phase 2
11. Consumer marketplace	Public discovery/booking app	Phase 3; do not build first

9. AI/ML strategy — practical, not buzzword-heavy Back to TOC

The prototype does not need a giant foundation model doing everything. Use deterministic optimization where the rules are known, ML where prediction is useful, and an LLM only for explanations or natural-language interaction.

Task	Recommended method	Why
Demand forecasting	LightGBM/XGBoost with lag features	Fast to train, explainable, works with

Task	Recommended method	Why
	+ calendar + weather/event signals; Prophet/ETS as baseline	modest data.
Prep optimization	Constrained optimization / linear programming / OR-Tools	Makes business constraints explicit: minimum display, batch size, shelf life, budget, service level.
Waste-risk score	Gradient boosting / probabilistic classifier	Produces an interpretable “risk of surplus” per item/time window.
Waste reason analysis	Rules + clustering; optional LLM summarization	Reasons are usually structured; LLM is useful for explaining patterns, not core computation.
Image recognition (later)	YOLO/RT-DETR or managed vision API + calibration scale	Useful for touchless waste capture, but hardware/process calibration matters.
Natural-language copilot	OpenAI/Gemini/Claude via tool calling	Answer: “Why did you cut paneer by 18?” using verified backend data; never let the LLM invent numbers.

10. Recommended tech stack Back to TOC

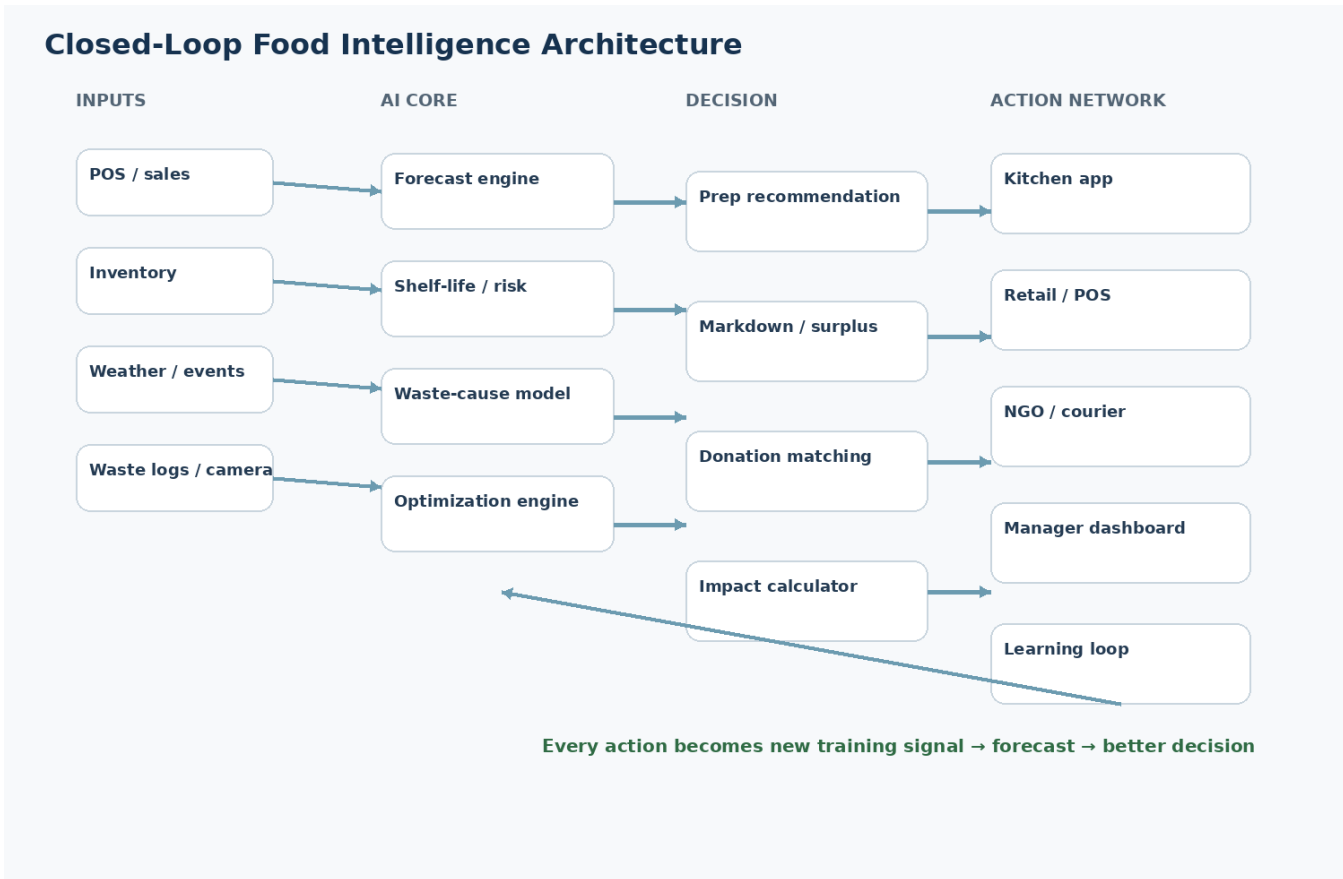
Layer	Choice	Reason
Frontend	Next.js + TypeScript + Tailwind + shadcn/ui + Recharts	Fast polished dashboard, responsive, production-friendly.
Mobile / PWA	Next.js PWA first; React Native later if required	One codebase for MVP; operators can use phones/tablets immediately.
Backend	FastAPI (Python) + Pydantic	Excellent for ML APIs, validation and async workflows.
Database	PostgreSQL	Transactional + analytical queries; strong multi-tenant foundation.
Cache / jobs	Redis + Celery/RQ or FastAPI background jobs	Forecast jobs, alerts and scheduled imports.
ML	scikit-learn + LightGBM/XGBoost + OR-Tools	Practical, explainable and feasible for student team.
LLM	OpenAI / Gemini / Claude	Explanation/coplay; swap provider through an abstraction layer.
Storage	S3/Azure Blob/GCP Storage	Images, invoices, datasets, audit

Layer	Choice	Reason
		artifacts.
Maps	Mapbox or Google Maps; OpenStreetMap as alternative	Nearby surplus partners and route visualization.
Auth	JWT/OAuth + RBAC	Business admin, manager, staff, NGO/courier roles.
Deployment	Vercel frontend + Render/Azure/AWS/GCP backend	Fast prototype deployment with clear upgrade path.

Avoid overengineering

Do not start with Kubernetes, microservices, custom vision hardware or blockchain. A modular monolith with clean domain boundaries is faster to build, easier to debug and completely adequate for an SIH prototype.

11. System architecture Back to TOC



11.1 Core data flow Back to TOC

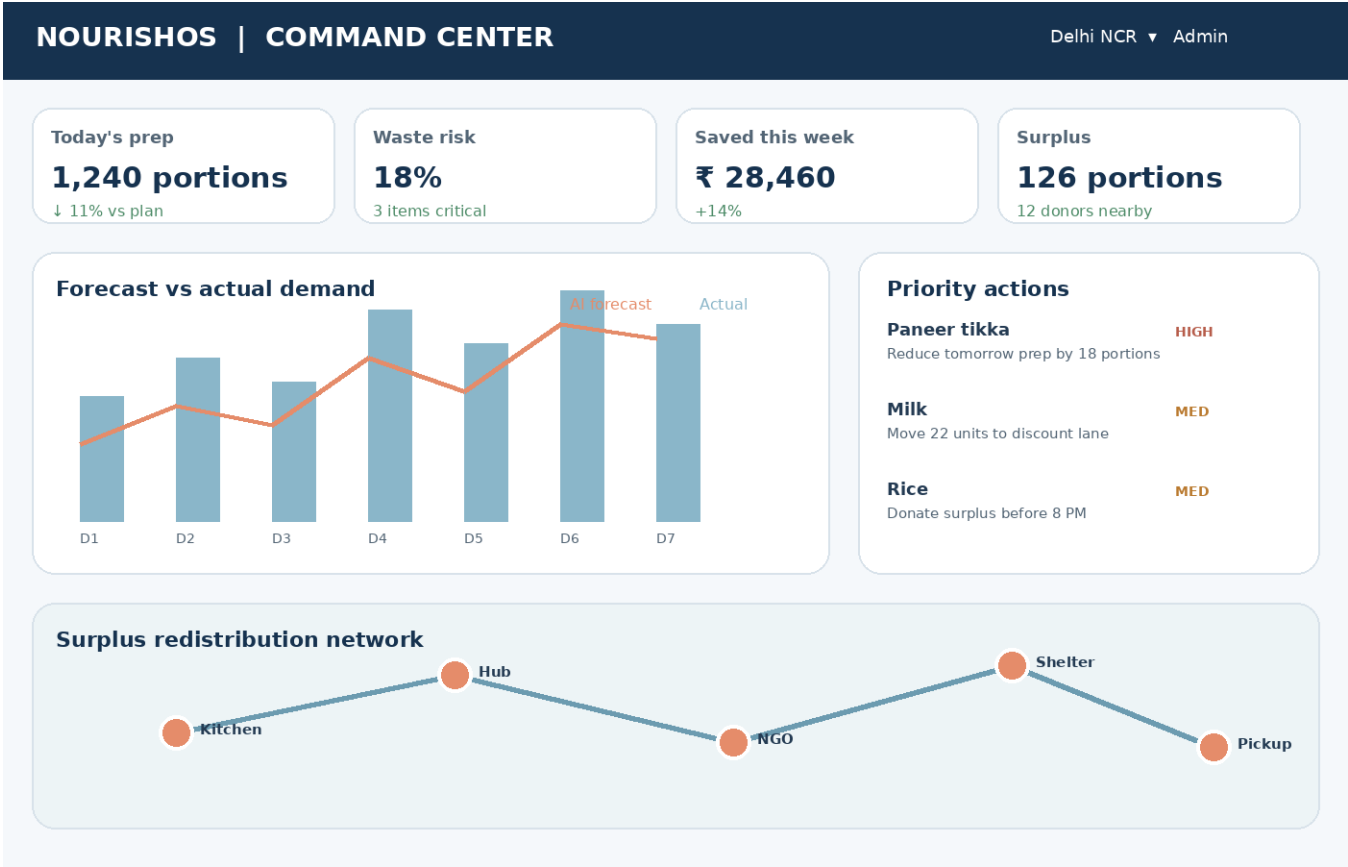
7. POS/inventory/menu/calendar/weather/event data enters ingestion layer.
8. Data quality service normalizes units, item IDs, recipes and timestamps.

- 9. Forecast service produces expected demand with uncertainty bounds.
- 10. Optimization service turns forecast into procurement and prep quantities.
- 11. Waste service records actual waste and labels causal stage/reason.
- 12. Surplus service evaluates sell/donate/recover options based on deadline and partner capacity.
- 13. Impact service calculates cost avoided and environmental proxies using versioned factors.
- 14. Feedback loop stores recommendation acceptance, override and outcome for model monitoring.

12. UI/UX direction — make the product feel like an operating system, not a sustainability portal

Back to TOC

The UI should communicate one idea: “Here is what is happening, here is what will likely happen next, and here is the best action you can take right now.” Avoid a wall of ESG charts. The first screen should prioritize decisions.



12.1 Core screens Back to TOC

Screen	Primary job	Key interaction
Command Center	See current risk and actions	Accept / modify / dismiss AI recommendation.
Forecast & Prep	Understand tomorrow/next-service demand	Drag batch quantity or click “simulate”.

Screen	Primary job	Key interaction
Waste Log	Capture an event in <10 seconds	Choose item → quantity → reason → done.
Waste Explorer	Find repeated causes	Filter by item, shift, reason, cost and location.
Surplus Hub	Route food before safety deadline	Compare sell / donate / recover paths and confirm pickup.
Impact	Tell the business what changed	View financial, food, and climate metrics with baseline comparison.

12.2 Mobile operator UX Back to TOC

The image displays three mobile app screens for waste management operations.

01 Log waste

What was wasted?

- Rice bowl: 2.4 kg
- Paneer: 1.1 kg
- Bread: 0.8 kg

Reason: Overproduction ▼

SAVE LOG

02 Fix tomorrow

Tomorrow's plan

- Breakfast: 168 portions
- Lunch: 245 portions
- Dinner: 212 portions

AI says: Cut Paneer Tikka by 18 portions

03 Route surplus

Surplus routing

126 portions
safe to redistribute

Sell
₹ 1,420 recovered

Donate
3 nearby NGOs

Compost
18 kg unavoidable

ROUTE NOW

UX rule #1

No staff member should need to type a paragraph while a kitchen is busy. Make the default workflow tap-first, voice-optional and barcode/scan-enabled later.

13. Interactivity — how the prototype should behave Back to TOC

Trigger	System response	Visible UI
Manager opens dashboard	Fetch latest risk + actions	Live KPI cards + prioritized action stack
Clicks “Why?”	Show evidence and contributing signals	Simple explanation: demand, weather, event, prior waste
Changes prep quantity	Recalculate expected cost/waste	Before vs after mini simulation
Logs waste	Update risk model & trend	Toast + updated waste driver rank
Surplus enters safe window	Search eligible channels	Sell / donate / recover cards with countdown
Action completed	Record outcome	Impact metric moves and action marked complete
Forecast was wrong	Show error + reason category	“Model missed by 18%; learning signal recorded”

14. Feasibility assessment Back to TOC

Dimension	Rating	Reality check
Technical feasibility	High	MVP can be built with standard web + Python ML stack and synthetic/CSV data.
Data availability	Medium	A production-grade model needs longitudinal POS/inventory data. Prototype can start with public/sample/simulated data and clear “demo data” labeling.
Hardware	Optional	Camera + scale is valuable but not required for MVP.
Integration	Medium	POS/ERP integrations are partner-specific; design APIs now, implement one CSV and one mock connector first.
Safety/compliance	Medium	Food safety must never be “predicted” by an LLM. Routing should respect configurable expiry,

Dimension	Rating	Reality check
		temperature and local policy rules.
Scalability	High	Multi-tenant PostgreSQL + async jobs + stateless API can scale substantially before architectural complexity is necessary.
Business viability	High potential	Clear ROI: lower procurement waste, better margins, fewer disposal costs, measurable sustainability reporting.

15. Business model and deployment model Back to TOC

Customer	Value proposition	Possible pricing model
Small restaurant/cafe	Simple prep + waste recommendations	Low monthly SaaS or freemium starter
Hotel / campus / hospital	Multi-outlet controls + waste analytics	Per site / per month
Grocery chain	Forecasting + markdown + inventory risk	Enterprise annual contract
NGO / food bank	Better surplus visibility and pickup planning	Free / subsidized network account
Government / city	Aggregate waste insights without exposing business details	B2G/B2B2G analytics contract

16. Globalization strategy Back to TOC

- Multi-currency, timezone, language, unit and measurement configuration from day one.
- Localization layer for food names, units, local food-safety rules and disposal/recovery pathways.
- Country-neutral core: demand, inventory, waste, routing and impact models are configurable domains.
- Open integration APIs so the platform can sit on top of existing POS/ERP systems rather than replacing them.
- Partner marketplace abstraction: the same surplus can route to a consumer marketplace, charity, community kitchen, animal-feed pathway or composting provider depending on local law and availability.
- Privacy by design: aggregate/tenant-isolated analytics, role-based access, encryption, audit logs and configurable retention.

Global principle

Do not build “an Indian food-waste app that can later be global.” Build a global core with Indian defaults. That means localizing the data—not rewriting the architecture.

17. What NOT to build in the first prototype Back to TOC

- An enormous consumer marketplace like Too Good To Go.
- Custom hardware manufacturing.
- Perfect food recognition for every cuisine in the world.
- Blockchain for traceability unless the official PS explicitly requires it.
- A giant chatbot replacing operational screens.
- AI-generated ESG numbers without source/version tracking.
- Dozens of dashboards with no action buttons.

18. The judge demo — 4 minutes Back to TOC

15. 0:00–0:30 — Show the waste problem as a business loss + climate issue; one number, one story.
16. 0:30–1:15 — Open Command Center: “Tomorrow’s demand is 510, not 550. Here is why.”
17. 1:15–2:00 — Change an external condition (occupancy/event/weather) and show the forecast/prep recommendation adapt.
18. 2:00–2:45 — Log a waste event and show the system identify the top repeated driver and recommend a corrective action.
19. 2:45–3:30 — Trigger surplus: system finds sell/donate/recover options, with countdown and route.
20. 3:30–4:00 — Show impact: cost avoided, kg food saved, portions recovered, trend improvement, and “learning signal recorded”.

The killer line

“We are not building another waste dashboard. We are building the decision loop that makes the dashboard improve tomorrow’s decision.”

19. Risks and how we handle them Back to TOC

Risk	Why it matters	Mitigation
Bad forecasts	A wrong recommendation can create stockouts	Show confidence, safe minimums and human override.
Garbage data	Inventory and SKU data are messy	Data-quality layer, item normalization and explicit uncertainty.
Overclaiming CO2	Impact factors vary by food and region	Version emissions factors; present as estimates with methodology.
Food safety	Unsafe redistribution is	Hard rules + expiry windows +

Risk	Why it matters	Mitigation
	unacceptable	verification; AI cannot override policy.
Low staff adoption	Busy workers skip complex tools	<10-second workflows and optional automation.
Integration complexity	Every POS is different	CSV first, adapter layer second, enterprise connectors later.
AI hallucination	LLM can fabricate explanations	LLM can only summarize structured backend evidence through tools.

20. Success metrics Back to TOC

Metric	Definition	Target for pilot
Avoidable waste rate	Avoidable waste / total food prepared	Downward trend vs baseline
Forecast MAPE/WAPE	Forecast error by item/time window	Improve weekly; track by category
Prep adherence	% recommendations accepted or intentionally overridden	>70% with reasons captured
Recovery rate	Surplus successfully sold/donated/recovered	Increasing trend
Cost avoided	Baseline waste cost – observed waste cost	Positive and attributable
Action completion	Recommendations completed before deadline	>60% in MVP pilot
Time to log	Median time for a waste entry	<10 sec mobile workflow

21. Prototype data model Back to TOC

Entity	Important fields
Organization	id, country, timezone, currency, business_type
Location	id, organization_id, address, operating_hours
Item	id, sku/plu, name, category, unit, shelf_life, cost, price
Recipe/MenuItem	id, item_id, ingredient_id, yield, portion_size

Entity	Important fields
DemandRecord	timestamp, location_id, item_id, quantity_sold, channel
InventorySnapshot	timestamp, item_id, on_hand, expiry_at, storage_zone
WasteEvent	timestamp, item_id, quantity, reason, stage, operator_id
Forecast	item_id, horizon, predicted_qty, lower_bound, upper_bound, model_version
Recommendation	type, item_id, action, rationale, expected_impact, confidence
SurplusLot	items, quantity, safe_until, status, origin_location
Partner	type, geography, capacity, pickup_window, verification_status
ImpactRecord	kg_saved, meals_recovered, cost_saved, co2e_estimate, factor_version

22. API sketch Back to TOC

Endpoint	Purpose
POST /v1/import/sales	Import sales history
POST /v1/import/inventory	Import inventory snapshot
GET /v1/forecast?location=...	Return demand forecast
POST /v1/optimize/prep	Calculate recommended prep
POST /v1/waste-events	Create waste event
GET /v1/insights/waste	Return ranked drivers
POST /v1/surplus/match	Find recovery options
POST /v1/recommendations/{id}/action	Accept/override recommendation
GET /v1/impact/summary	Return impact dashboard

23. Implementation plan Back to TOC

Phase	Build	Exit criterion
Phase 0	Dataset + product rules + UI skeleton	Can demo one full day of synthetic business data.
Phase 1	Forecast + prep optimizer + dashboard	Forecast changes when inputs change.
Phase 2	Waste logging + root-cause insights	Waste event updates dashboard immediately.
Phase 3	Surplus routing	At least three route types with deadline logic.
Phase 4	Polish + storytelling	4-minute judge demo is smooth, visual and reproducible.
Phase 5	Optional wow	Computer vision waste capture, voice interface, simulated multi-site network.

24. Final recommendation Back to TOC

Build this as a “Food Operations Intelligence OS,” not a waste tracker. The first product promise should be: “Prepare closer to demand, see waste before it repeats, and recover what remains.”

For SIH, the best balance is a credible MVP with real calculations and interactive flows rather than a huge platform full of fake integrations. Show one business location in depth, make the system visibly react to changes, and make every recommendation explainable.

What would make this genuinely stand out

The strongest version combines three kinds of intelligence in one visible loop: Predictive AI (what will happen), Optimization (what should we do), and Network Intelligence (what should happen to unavoidable surplus). Most competitors are strongest in one or two of these layers; your prototype can make the integration itself the innovation.

25. Research sources Back to TOC

[0] SIH 2026 Problem Statements Overview (third-party indexed copy; used only to flag public-source ambiguity) — <https://www.scribd.com/document/913856423/Problem-Statements-Smart-India-Hackathon-Sih>

[1] UNEP — Food Waste Index Report 2024 — <https://www.unep.org/resources/publication/food-waste-index-report-2024>

[2] UNEP — World squanders over 1 billion meals a day — <https://www.unep.org/news-and-stories/press-release/world-squanders-over-1-billion-meals-day-un-report>

[3] FAO — Food Loss and Food Waste policy gateway — <https://www.fao.org/policy-support/policy-themes/food-loss-and-food-waste/en>

- [4] Winnow — AI Food Waste Prevention System — <https://www.winnowsolutions.com/>
- [5] Orbisk — Automatically reduce food waste in your kitchen — <https://orbisk.com/>
- [6] Orbisk — AI-powered Actions — <https://orbisk.com/product/orbisk-ai/>
- [7] Leanpath — AI Food Waste Solutions — <https://www.leanpath.com/>
- [8] Leanpath — Food Waste Management Solutions — <https://www.leanpath.com/solutions/>
- [9] Too Good To Go — Platform — <https://www.toogoodtogo.com/en-us/platform>
- [10] FoodCloud — Surplus Food Redistribution — <https://www.food.cloud/>
- [11] Wasteless — AI-powered Markdowns — <https://www.wasteless.com/ai-markdowns>
- [12] Freshflow — AI for fresh produce ordering — <https://www.freshflow.ai/>
- [13] Afresh — Grocery AI platform — <https://www.afresh.com/platform>